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Supercharged Summers: Understanding 2022's Climate Extremes

Abstract

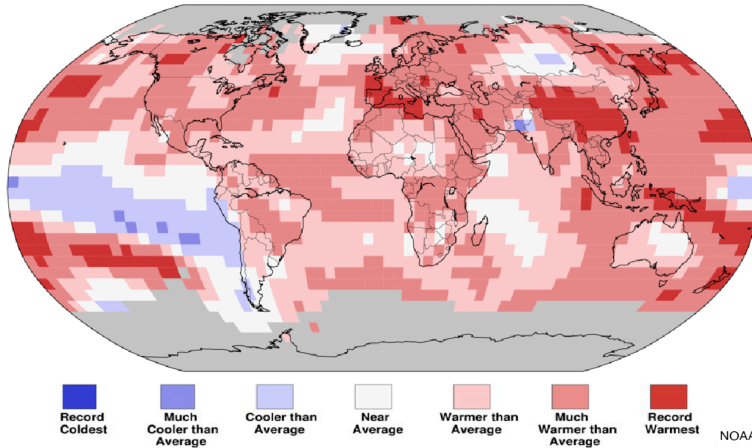
The Northern Hemisphere summer of 2022 brought disaster after disaster: record heat in India and Pakistan, floods covering a third of Pakistan, drought and wildfires in Europe, and rivers running dry in China and the U.S. Climate scientist Kevin Trenberth explained how global warming accelerates the water cycle, making wet areas wetter and dry areas drier, and how La Niña shaped where extremes struck.

Introduction

There is an old joke about a man with one foot in ice water and the other in hot water, so his average temperature is fine. Summer 2022 felt like that. June to August 2022 tied as the fifth-warmest on record globally, 1.6°F above the 20th-century average, but the averages hid extremes. Temperatures broke 122°F (50°C) in Jacobabad, Pakistan. Monsoon floods then left about a third of Pakistan under water, affecting some 33 million people. Europe baked, China's Yangtze River fell to its lowest level since at least 1865, and Lake Powell and Lake Mead hit record lows. How can climate change cause both floods and droughts?

How Do Scientists Study This?

Surface temperature percentiles for June-August 2022

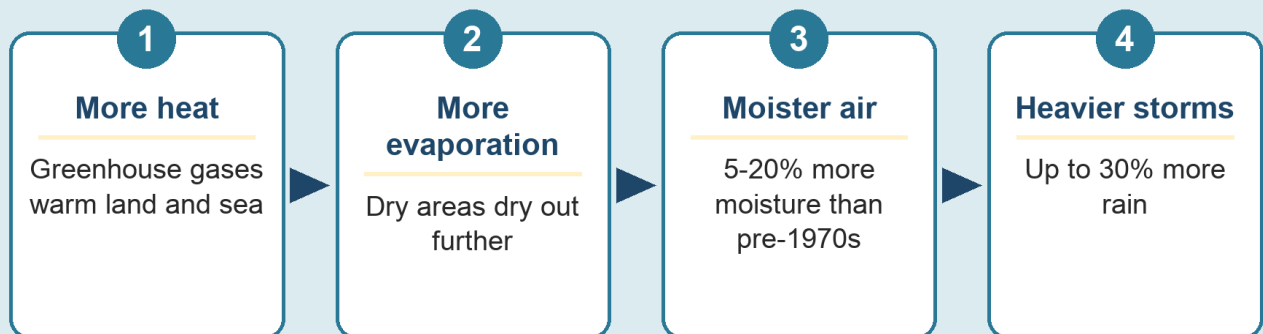


FROM THE ARTICLE

Surface temperature percentiles for June-August 2022. Image: NOAA.

Trenberth explained the pattern through the physics of the water cycle. Warming increases evaporation, drying areas with little rain. Warmer air can hold more water vapor, so a thirstier atmosphere pulls moisture from the surface. Winds carry that moisture into storms, often a thousand miles away, where it falls as heavier rain. Because precipitation falls on only about 8% of the land at any one time, this produces exaggerated extremes: localized downpours and widespread dry spells. He then added natural variability, examining how a third year of La Niña, with cool tropical Pacific waters and marine heat waves in the North and South Pacific, steered where the extremes occurred.

How warming supercharges the water cycle



Study the Process

Follow the water

Trace how warming increases evaporation, drying some regions and moistening the air.

Find the downpours

Extra moisture flows into distant storms, which rain harder and longer.

Add natural patterns

La Niña and El Niño decide where the wet and dry extremes land.

What Did They Discover?

Atmospheric moisture has increased 5% to 20% compared with before the 1970s, and storms can produce up to 30% more rain. With an accelerated water cycle, wet areas get wetter and dry areas drier. In Pakistan, three times the normal annual rain fell during the monsoon; more than 1,500 people died and about 1.8 million homes were damaged or destroyed. Drought in Spain revealed an ancient stone circle, low rivers cut French nuclear power output, and flash floods in eastern Kentucky killed more than three dozen people. Whether rain becomes a disaster depends on drainage, where people live, and whether infrastructure was built for today's climate.

Summer 2022 by the numbers

33M

people affected by Pakistan's floods

3×

normal annual rainfall during Pakistan's monsoon

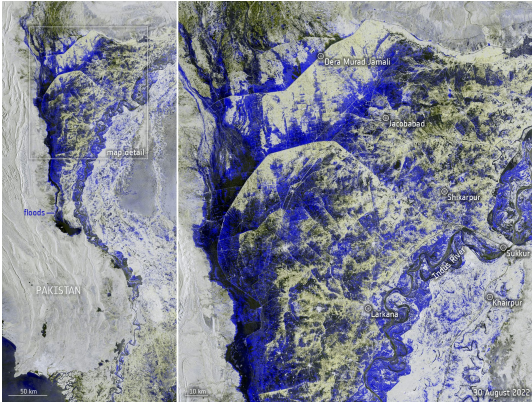
122°F

reached in Jacobabad, Pakistan, in May

Trenberth warned that exceptionally warm deep water in the western Pacific suggested an El Niño in 2023, potentially bringing more global temperature records in 2024.

Look at the Evidence

FROM THE ARTICLE



Satellite images of Pakistan show flooding turning rivers into lakes several miles wide. Image: European Space Agency.



Residents fought wildfires in Spain in July 2022 that spread through dry fields and forests. Photo: AP Photo/Bernat Armangué.

Takeaways

Wet gets wetter

A warmer atmosphere carries more moisture into storms, fueling heavier rain.

Dry gets drier

Warming speeds evaporation, deepening droughts where rain is scarce.

Nature decides where

La Niña and El Niño shift which regions face floods or drought.

Key Terms

Water cycle

The continuous movement of water through evaporation, condensation, precipitation, and runoff.

Monsoon

A seasonal shift in winds that brings heavy rains to regions such as South Asia.

Latent heat

Heat absorbed when water evaporates and released when it condenses, which fuels storms.

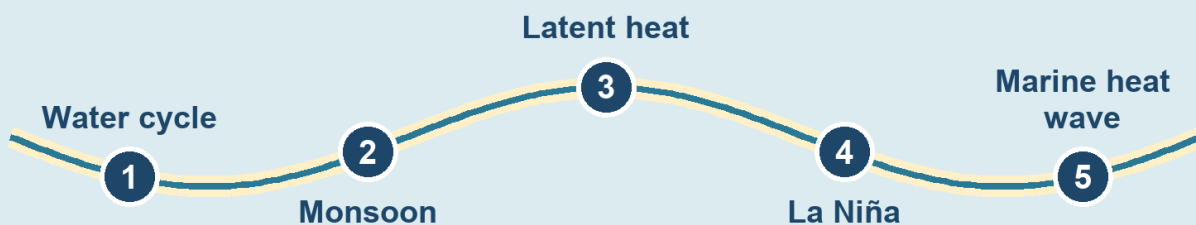
La Niña

A natural pattern of cooler waters in the central and eastern tropical Pacific that shifts weather worldwide.

Marine heat wave

A period of unusually high ocean temperatures in a region.

Vocabulary Path · Supercharged Summers



Activity

Disaster Map of 2022

Students map 2022's climate extremes and explain each one using the accelerated water cycle.

Materials: World map; Event cards: Pakistan floods, Europe drought and fires, Yangtze drought, Kentucky floods, Lake Mead low; Two colors of markers

1. Place each event on the map, coloring floods in blue and droughts or heat in red.
2. For each event, write one number from the lesson that shows its severity.
3. Use the idea that wet gets wetter and dry gets drier to explain two events.
4. Add the La Niña pattern and discuss how it influenced where extremes occurred.
5. Choose one event and propose how better infrastructure could have reduced the damage.

References

Trenberth, K. (2022). 2022's supercharged summer of climate extremes: How global warming and La Niña fueled disasters on top of disasters. The Conversation.
<https://theconversation.com/2022s-supercharged-summer-of-climate-extremes-how-global-warming-and-la-nina-fueled-disasters-on-top-of-disasters-190546>